

GRADE 9 BIOLOGY • CHAPTER 2 NOTES

Cell: The Building Block of Life

Topics Covered: Origin of Life • Microscopy & Cell Size Estimation • Cell Boundary (Membrane & Wall) • Transport (Osmosis & Diffusion) • Prokaryotes vs. Eukaryotes • Organelles • Cell Division (Mitosis & Meiosis) • Cell Theory & Cancer • Complete Solved Exercises

CENTRAL QUESTIONS (THINK IT OVER)

- Where does a cell come from?
- How have technological interventions facilitated the creation of new knowledge beyond the naked eye?
- How is the cell the fundamental structural and functional unit of life?
- How do cells grow and multiply?

1. Introduction & Origin of Life

- **Origin of Life:** Scientifically accepted that life originated in water ~3.5 billion years ago, likely in hot springs (e.g., Puga Valley in Ladakh, India, where temperatures reach near boiling point). Ancient thermophilic (heat-loving) unicellular bacteria survived under these conditions.
- **Protective Membrane Formation:** Studies from Birbal Sahni Institute of Palaeosciences (Lucknow) revealed that rapid calcium carbonate precipitation around hot springs protected organic molecules from radiation and aided in forming the first protective membrane.
- **Structural Organization Hierarchy:**
Cells → Tissues → Organs → Organ Systems → Organism
 (e.g., Nasal pores + Nasal cavity + Trachea + Lungs = Respiratory System).
- **Unicellular vs. Multicellular:** Unicellular organisms (e.g., bacteria, yeast, Amoeba) consist of a single cell. Multicellular organisms (e.g., plants, fish, humans) consist of millions of coordinated cells.

2. How to Study Cells? (Microscopy & Scale)

Resolution & Visibility Limits

- **Limit of Resolution of Human Eye:** The ability to distinguish two close points as separate. From ~25 cm (near point), human eyes can distinguish points separated by **0.1 mm**. Structures smaller than 0.1 mm appear merged.
- **Light Microscope:** Uses visible light and glass convex lenses (eyepiece + objective lens). Magnifies up to ~1000X–1500X. Robert Hooke first discovered box-like compartments in cork in 1665 using a self-designed microscope (200–300X).
- **Electron Microscope:** Uses electron beams instead of light to achieve nanometer-scale ($1 \text{ nm} = 10^{-9} \text{ m}$) resolution with remarkable clarity. Reveals fine ultrastructure (stomata, organelle membranes).

Size Scale of Natural & Biological Objects

Scale Range	Objects / Biological Entities	Observation Tool
100 m – 1 m	Neem tree, Rocket, Human height, Great Indian Bustard	Unaided Eye
0.1 m – 1 mm	Chicken egg, Fish egg, Amoeba	Unaided Eye / Magnifying Glass
100 μm – 1 μm	Plant & Animal cells (10–100 μm), Nucleus, Bacteria (1–10 μm), Mitochondria	Light Microscope
100 nm – 0.1 nm	Smallest bacteria, Viruses (100 nm), Ribosomes (10 nm), Proteins, Lipids, Atoms (0.1 nm)	Electron Microscope

ACTIVITY 2.1: ESTIMATING CELL SIZE & MAGNIFICATION

1. Unit Conversion: $1 \text{ mm} = 1000 \text{ } \mu\text{m}$.

2. Formula for Real Cell Size:

$$\text{Estimated Cell Size} = \frac{\text{Diameter of field of view in } \mu\text{m}}{\text{Number of cells lined up across the diameter}}$$

Example: If visible field = $5 \text{ mm} = 5000 \mu\text{m}$ and 25 onion cells span the diameter, then individual cell size = $5000 / 25 = 200 \mu\text{m}$.

3. Total Magnification: $\text{Total Magnification} = \text{Power of Eyepiece} \times \text{Power of Objective Lens}$. (e.g., $10 \times 10 = 100$).

3. The Cell Boundary & Transport Mechanisms

A. Cell Membrane (Plasma Membrane)

- **Universal Feature:** Thin (7–10 nm) protective boundary surrounding all living cells.
- **Selectively Permeable:** Allows specific substances (water, gases) to pass while blocking others.
- **Fluid-Mosaic Model:**
 - *Lipid Bilayer:* Double layer of fat molecules with water-attracting (hydrophilic) heads pointing outwards and water-repelling (hydrophobic) tails pointing inwards.
 - *Fluidity:* Lipid and protein molecules can rotate, flip, and move laterally.
 - *Mosaic Pattern:* Embedded proteins act as gatekeepers/channels for transport.

B. Diffusion vs. Osmosis

- **Diffusion:** Net movement of particles from higher concentration to lower concentration along a concentration gradient (occurs without needing a membrane).
- **Osmosis:** Net movement of water molecules across a **selectively permeable membrane** from a dilute solution (higher water/lower solute) to a concentrated solution (lower water/higher solute).

C. Effect of Solutions on Cells (Tonicity)

Solution Type	Solute Concentration Comparison	Net Water Movement	Effect on Animal Cell	Effect on Plant Cell
Isotonic	Extracellular = Intracellular	No net movement	Shape remains same	Normal / Flaccid
Hypotonic	Extracellular < Intracellular (dilute)	Water enters cell	Cell swells & may burst	Cell becomes turgid (firm)
Hypertonic	Extracellular > Intracellular (concentrated)	Water leaves cell	Cell shrinks (crenation)	Plasmolysis (contents shrink away from cell wall)

ACTIVITY 2.2: POTATO OSMOSIS & ACTIVITY 2.3: PLANT VS. ANIMAL CELLS

Potato Osmosis: Potato in plain water (Beaker A) swells due to endosmosis. Potato in 20% salt solution (Beaker B) shrinks due to exosmosis.

Cell Observation: Onion peel / Rhoeo cells are regularly arranged box-shaped cells with rigid borders. Cheek cells are irregular thin-walled cells.

Plasmolysis in Rhoeo: In 20% sugar solution, the cell wall retains its rectangular shape, while the inner plasma membrane and cytoplasm shrink away from the wall.

D. Cell Wall

- **Location & Composition:** Present outside plasma membrane in plants (made of **cellulose**), fungi (chitin), and bacteria. Absent in animal cells.
- **Permeability:** Fully permeable to water and dissolved minerals.
- **Function:** Provides structural rigidity to withstand wind/rain, prevents cell bursting in hypotonic environments, and maintains upright plant posture. Dietary cellulose provides roughage for human digestion.

4. Cell Interior: Prokaryotes vs. Eukaryotes

Prokaryotic vs. Eukaryotic Cell Comparison

Characteristic	Prokaryotic Cell	Eukaryotic Cell
Nucleus	Primitive nucleoid (single circular DNA without membrane)	Well-defined nucleus surrounded by double nuclear membrane
Diameter	1 to 10 μm	10 to 100 μm
Organisms	Unicellular (Bacteria, Archaea)	Unicellular or Multicellular (Plants, Animals, Fungi)
Membrane-Bound Organelles	Absent (activities occur directly in cytoplasm)	Present (ER, Golgi, Mitochondria, Plastids, Lysosomes)
Cytoskeleton	Absent / Primitive	Present (network of fine protein fibers for support & transport)

ACELLULAR INFECTIOUS AGENTS (BEYOND CELLS)

Viruses, viroids, and prions lack cellular structure.

- **Viruses:** Genetic material (DNA/RNA) enclosed in a protein coat.
- **Viroids:** Infectious RNA lacking a protein coat.
- **Prions:** Misfolded infectious proteins lacking genetic material.

5. Detailed Structure & Function of Eukaryotic Organelles

1. Nucleus (House of Coded Instructions)

- **Structure:** Enclosed by a double-layered nuclear membrane containing nuclear pores for nucleocytoplasmic transport. Contains a dense spherical body called the **nucleolus** (site of ribosomal subunit assembly).
- **Genetic Material:**
 - *Chromatin:* Entangled thread-like mass of DNA and proteins in non-dividing cells.
 - *Chromosomes:* Rod-shaped structures formed when chromatin condenses during cell division.
 - *Genes:* Functional segments of DNA containing heredity instructions.
- **Enucleated Specialization:** Mature human Red Blood Cells (RBCs) lack a nucleus to maximize space for hemoglobin (oxygen transport), resulting in a 120-day lifespan without self-repair.

2. Ribosomes (Protein Factories)

- Tiny non-membrane-bound structures made of two subunits (assembled in nucleolus). Found floating free in cytoplasm or attached to RER. Primary sites of protein synthesis.

3. Endoplasmic Reticulum (ER - Manufacturing Network)

- Extensive membrane network continuous with the outer nuclear membrane.
 - **Rough ER (RER):** Studded with ribosomes; synthesizes and secretes proteins (e.g., pancreatic gland cells).
 - **Smooth ER (SER):** Lacks ribosomes; synthesizes and stores lipids, fats, and steroid hormones.

4. Golgi Apparatus (Packaging & Shipping Center)

- Stacks of flattened sac-like cisternae discovered in 1898 by Camillo Golgi in nerve cells of barn owls.
- Functionally linked to ER: Modifies, sorts, and packages proteins/lipids into vesicles for secretion, membrane repair, or lysosome formation.

5. Lysosomes (Clean-Up System)

- Single-membrane sacs filled with digestive hydrolytic enzymes. Break down waste, worn-out organelles, and foreign material.
- *Special Role:* Human sperm heads contain lysosomal enzymes (acrosome) to digest the outer egg layer during fertilization.

6. Mitochondria (Powerhouse of the Cell)

- Double-membrane organelle. Outer membrane is smooth/porous; inner membrane is folded into **cristae** to increase surface area for ATP production via cellular respiration.
- Generates **ATP (Adenosine Triphosphate)**, the energy currency of the cell. Contains its own circular DNA and 70S-like ribosomes (can make its own proteins).

7. Plastids (Food Synthesis & Storage - Plants)

- Double-membrane organelles present in plants and algae. Have their own DNA and ribosomes.
 - **Chloroplasts:** Contain green pigment *chlorophyll*. Semi-fluid *stroma* contains disc-shaped thylakoid membranes for light absorption in photosynthesis.
 - **Chromoplasts:** Contain non-green pigments (yellow, orange, red) in flowers/fruits to attract pollinators and fruit eaters.
 - **Leucoplasts:** Colourless storage plastids (store starch in potatoes/taro, oils, or proteins).

8. Vacuoles (Storage & Turgidity)

- **Plant Cells:** Single large central vacuole filled with cell sap (water, minerals, sugars, waste), surrounded by the *tonoplast* membrane. Maintains turgor pressure (prevents wilting).
- **Animal Cells:** Small, temporary vacuoles for transport or waste storage.

6. How Normal Cells Grow and Divide

Cell growth occurs not by individual cells expanding infinitely, but through **cell division**. Humans replace hundreds of billions of cells daily (~1% of total body cells).

ACTIVITY 2.5: OBSERVING MITOSIS IN ONION ROOT TIP

Fresh onion roots grown in water are fixed in aceto-alcohol (1:3 glacial acetic acid:ethanol), softened in dilute HCl, stained with aceto-carmin, squashed, and observed under a microscope to view continuous stages of nuclear and cytoplasmic division.

Types of Cell Division

Feature	Mitosis	Meiosis
Primary Purpose	Growth, tissue repair, maintenance, asexual reproduction	Formation of gametes for sexual reproduction (genetic diversity)
Occurrence Location	Somatic (body) cells (e.g., skin, root tip)	Reproductive organs (Testes/Ovaries; Anthers/Ovaries)
Number of Divisions	One single division	Two sequential divisions (Meiosis I & Meiosis II)
Daughter Cells Produced	2 genetically identical daughter cells	4 genetically diverse daughter cells
Chromosome Number	Diploid (2n) → Same as parent	Haploid (n) → Reduced to half

Biotechnology & Medical Insights

- **Cell Culture:** Growing isolated cells in sterile nutrient media under regulated pH, temperature, and humidity for manufacturing vaccines, biochemicals, and drugs.
- **Mitosis Errors & Cancer:** Loss of **contact inhibition** causes uncontrolled mitotic division, forming benign or malignant *tumours*.
- **Meiosis Errors:** Leads to abnormal chromosome numbers, developmental genetic disorders, early pregnancy loss, or reduced fertility.
- **Programmed Cell Death (PCD / Apoptosis):** Genetically regulated cell self-destruction required for digit separation during embryonic development (prevents webbed hands) and tissue quality control.
- **Totipotency (Gottlieb Haberlandt, 1902):** Ability of any single living plant cell to develop into an entire plant when given proper nutrients, establishing *Plant Tissue Culture Technology*.
- **Synthetic Biology (J. Craig Venter, 2010):** Synthesized the genome of *Mycoplasma mycoides* electronically, synthesized it in a lab, and transplanted it into an emptied bacterial cell, proving DNA controls cell activity.

7. Cell Theory & Historical Discoveries

- **Matthias Schleiden (1838):** Reported that all plants are composed of cells.
- **Theodor Schwann (1839):** Reported that all animals are composed of cells.
- **Rudolf Virchow (1855):** Added "*Omnis cellula-e-cellula*" — All new cells arise from pre-existing cells.
- **Classical Cell Theory Principles:**
 1. All living organisms are composed of one or more cells.
 2. The cell is the basic structural and functional unit of life.
 3. All cells arise from pre-existing cells through division.

8. Complete Chapter Exercise Solutions (Revise, Reflect, Refine)

Q1. Differentiate between pairs based on clues:

- (i) **Cell membrane vs Cell wall (Permeability):** Cell membrane is *selectively permeable* (controls specific entry/exit), whereas Cell wall is *fully permeable* to water and dissolved solutes.
- (ii) **RER vs SER (Structure):** RER has a *rough surface studded with ribosomes*, whereas SER has a *smooth tubular surface lacking ribosomes*.
- (iii) **Chloroplasts vs Chromoplasts (Pigments):** Chloroplasts contain green pigment *chlorophyll*, whereas Chromoplasts contain yellow, orange, or red carotenoid pigments.

Q2. Animal cell X in pure water swells; Cell Y in concentrated salt shrinks. Explanation?

Correct Option: (iii) Water moved into Cell X and moved out of Cell Y through the cell membrane.

Reason: Pure water is hypotonic to Cell X (endosmosis causing swelling). Salt solution is hypertonic to Cell Y (exosmosis causing shrinkage).

Q3. Match labelled parts of Plant Cell diagram (Fig 2.20) with functions:

- **(a) Mitochondria:** (ii) Site of cellular respiration.
- **(b) Nucleus:** (i) Controlling all the activities of a cell.
- **(c) Golgi apparatus:** (vi) Packs and stores materials received from ER.
- **(d) Chloroplast:** (vii) Helps in manufacturing food.
- **(e) Cell wall:** (v) Provides structural rigidity to the cell.
- **(f) Cell membrane:** (iv) Separates the cell contents from surroundings.
- **(g) Vacuole:** (iii) Storage organelle that also provides rigidity to the cell.

Q4. Pairs of organelles correctly placed under given categories:

Correct Option: (i) Present in plant cells: Leucoplast | Absent in animal cells: Cell wall.

Q5. Renu states all plant parts including roots contain plastids. Rohit claims roots lack plastids as they don't photosynthesize. Who is correct?

Renu is correct. Plant roots contain **leucoplasts** (a type of non-pigmented plastid) which function to store food materials like starch, proteins, and lipids (e.g., starch storage in potato tubers and carrot roots). While root cells lack chloroplasts due to lack of sunlight, plastids as a general organelle family are present.

Q6. Similarities and differences between Mitochondria and Chloroplasts.

Similarities: Both are double-membrane organelles, contain their own circular DNA and ribosomes, can synthesize some of their own proteins, and are involved in energy transformations.

Differences: Mitochondria perform cellular respiration (break down glucose to ATP) and are present in both plant and animal cells. Chloroplasts perform photosynthesis (synthesize glucose using light) and are present only in plant/algal cells.

Q7. Which pair of organelles contains DNA?

Correct Option: (ii) Mitochondria, Nucleus. (Chloroplasts also contain DNA).

Q8. Carrot experiment (Plain water vs Concentrated salt solution):

(i) *Hypothesis*: To test the effect of osmotic water movement on plant tissue firmness in solutions of different solute concentrations.

(ii) *Improvement*: Measure initial and final weights and dimensions of the carrots before and after 24 hours using a precise weighing balance.

(iii) *Explanation*: In plain water (hypotonic), water enters carrot cells via endosmosis, creating turgor pressure that keeps cells stiff and crunchy. In salt solution (hypertonic), water leaves cells via exosmosis, causing loss of turgor (plasmolysis) so the carrot becomes limp and rubbery.

Q9. Presence/Absence table for Bacterial and Animal Cells:

Structure	Bacterial Cell	Animal Cell
Chromosome	Present (Single circular nucleoid)	Present (Multiple linear in nucleus)
Nucleus	Absent	Present
Mitochondria	Absent	Present
Golgi complex	Absent	Present
Chromoplasts	Absent	Absent

Q10. Four potato cups experiment (A: empty, B: sugar, C: salt, D: boiled + sugar in water beaker):

(i) *Water in B & C*: Sugar/salt inside the cup creates a hypertonic environment relative to the water outside. Water moves through living potato cells by osmosis into the hollowed cavity.

(ii) *Role of Cup A*: Serves as a **control experiment** to prove that water movement requires a concentration gradient and does not accumulate spontaneously.

(iii) *Why no water in A & D*: Cup A lacks solute gradient. In Cup D, boiling kills potato cells and destroys selective permeability of cell membranes, preventing osmosis.

Q11. Incorrectly matched organelle-function pair:

Incorrect Pair: (ii) SER — Lipid and cellulose synthesis.

Correction: SER synthesizes lipids and steroids, but *cellulose is synthesized at the plasma membrane by cellulose synthase enzyme complexes*.

Q12. What outcome do you expect if all mitochondria are removed from a eukaryotic cell?

The cell will be unable to generate ATP through aerobic cellular respiration. Essential energy-dependent processes (active transport, protein synthesis, movement) will cease, leading to cellular failure and cell death.

Q13. Which phenomenon inhibits tumor formation? Can plants develop tumors?

Contact Inhibition inhibits tumor formation in animal cells. Plants *can also develop tumors* (e.g., galls caused by *Agrobacterium tumefaciens* infection), but plant cells do not display contact inhibition due to their rigid cell walls.

Q14. Synthesis and transport path of cell membrane lipids and proteins:

Proteins are synthesized at **RER** and lipids at **SER** → Packaged into transport vesicles → Shipped to **Golgi Apparatus** (modified/sorted) → Secretory vesicles fusion with **Plasma Membrane** (membrane biogenesis).

Q15. What would happen if gametes were formed by mitotic divisions?

Gametes would remain diploid ($2n$). Upon fertilization, the offspring chromosome number would double every generation ($4n$, $8n$...), disrupting genetic stability and cellular functioning.

Q16. Farmer Deepa preservation case study (Pickles/Sharbat):

(i) *Scientific Concept*: **Osmosis / Exosmosis**.

(ii) *Action of High Salt/Sugar*: Creates a hypertonic environment. Spoilage bacteria/fungi lose water via exosmosis, undergo plasmolysis, and cannot survive or multiply.

(iii) *Healthy Recipe*: Lemon-ginger sharbat preserved using jaggery/honey syrup or amla pickle in rock salt.

(iv) *Scientific Values*: Application of science to reduce food waste, value addition, sustainable agriculture, and strengthening local food security.